

Action–Angle Components, Exact Return Orders, and Zeta Laws for Every Periodic Box–Ball Level Set

Route-A structural certificate C182 — Round 1

Abstract

For the periodic binary box–ball system, we freeze every $L \geq 2M$, every soliton content and internal-symmetry sector, and every commuting carrier evolution T_l . The action–angle bijection and invariant-torus decomposition are prior theorems of Kuniba–Takagi–Takenouchi and Takagi, not novelty claims. From their precise component matrix we derive a single all-parameter theorem: augmented Smith normal form gives the exact order of each T_l translation; component formulas aggregate over symmetries, contents, fixed mass, and the full positive-weight space; Möbius inversion gives primitive cycles; and the Artin–Mazur zeta is the reciprocal ordinary determinant of the finite counting-measure Koopman permutation. Vacuum, half filling, and saturated carrier capacity are included. An independent periodic-carrier enumeration checks every binary state through $L = 9$. The exact cycles have no intrinsic rational-prime origin or logarithmic arithmetic clock, so the candidate is mathematically retained and Route-A rejected.

Keywords: periodic box–ball system; action–angle variables; internal symmetry; Smith normal form; dynamical zeta; Koopman permutation.

中文摘要

本文固定周期二元盒球系统的全部参数： $L \geq 2M$ 、任意孤子含量、任意内部对称扇区以及全部可交换载流子演化 T_l 。作用角变量双射与不变环面分解属于 Kuniba–Takagi–Takenouchi 和 Takagi 的既有定理，本文不声称其新颖性。在该定理上，我们用增广 Smith 标准形给出每个平移的精确阶，把不动点公式逐层汇总到全部对称、全部孤子含量、固定球数和完整正权状态空间；再由莫比乌斯反演得到本原周期，并证明动力 ζ 函数等于有限计数测度 Koopman 置换普通行列式的倒数。真空、半填充和载流子饱和和边界均被保留。独立的周期载流子程序穷举核验了 $L \leq 9$ 的所有二元状态。由于这些精确周期没有内生素数来源或对数算术时钟，数学结果被保留，但 Route-A 候选被否决。

关键词：周期盒球系统；作用角变量；内部对称；Smith 标准形；动力 ζ 函数；Koopman 置换。

1 Frozen source dynamics and action data

Let $\mathcal{P}_{L,M}$ be the binary words on a ring of length L with M balls, where $0 \leq M \leq \lfloor L/2 \rfloor$. The capacity- l periodic combinatorial- R carrier defines T_l ; one application is one unit of the discrete physical clock. KTT prove that the T_l commute and preserve the soliton content [1]

$$m = (m_j)_{j \geq 1}, \quad \sum_{j \geq 1} j m_j = M.$$

Put $H = \{j : m_j > 0\}$ and

$$p_j = L - 2 \sum_{k \in H} \min(j, k) m_k. \quad (1)$$

Since $L \geq 2M$, every $p_j \geq 0$; for $N = \max H$, $p_N = L - 2M$. Thus equality in the density bound is visible as $p_N = 0$, rather than excluded.

For every $j \in H$, choose

$$\alpha_j \mid \gcd(m_j, p_j), \quad \gcd(m, 0) = m,$$

and order H increasingly. Define

$$F_{jk}^{(\alpha)} = \frac{\delta_{jk} p_k + 2 \min(j, k) m_k}{\alpha_k}, \quad h_l = (\min(j, l))_{j \in H}. \quad (2)$$

The KTT/Takagi component theorem states that every connected component of the symmetry- α level is equivariantly bijective to [2]

$$G_\alpha = \mathbb{Z}^H / F^{(\alpha)} \mathbb{Z}^H, \quad T_l[x] = [x + h_l]. \quad (3)$$

It has $D_\alpha = \det F^{(\alpha)} > 0$ points.

The exact number of copies of (3) is

$$c(\alpha) = \prod_{j \in H} \frac{|\Lambda^{(\alpha_j)}(m_j, p_j)|}{m_j / \alpha_j}, \quad (4)$$

where

$$|\Lambda^{(\alpha)}(m, p)| = \sum_{\substack{\beta | \gcd(m, p) \\ \alpha | \beta}} \mu(\beta / \alpha) \binom{(p + m) / \beta - 1}{m / \beta - 1}. \quad (5)$$

Equation (5) is Möbius inversion by exact internal period. The denominator in (4) is integral because cyclic shift acts freely with orbit size m / α on an exact-period sequence. The sector identity is

$$\Omega_L(m) = \sum_{\alpha} c(\alpha) D_{\alpha} = \det F^{(1)} \prod_{j \in H} \frac{1}{m_j} \binom{p_j + m_j - 1}{m_j - 1}. \quad (6)$$

These statements are imported with attribution. Our contribution starts from the exact finite translations in (3) and derives their complete fixed-cycle-determinant consequences.

2 Augmented Smith order

Fix one component, abbreviate $F = F^{(\alpha)}$, $h = h_l$, and $D = \det F$, and let $B = [F \mid h]$. Write the nonzero Smith factors of B as $e_1, \dots, e_s$.

Theorem 1 (Exact component return order). *The order of T_l on the component is*

$$q_{\alpha, l} = \min\{n \geq 1 : nh_l \in F^{(\alpha)} \mathbb{Z}^H\} = \frac{D_{\alpha}}{e_1 \cdots e_s}. \quad (7)$$

If $UFV = \text{diag}(d_1, \dots, d_s)$, equivalently

$$q_{\alpha, l} = \text{lcm}_i \frac{d_i}{\gcd(d_i, (Uh_l)_i)}.$$

Indeed, quotienting G_{α} by the cyclic subgroup generated by $[h]$ gives

$$G_{\alpha} / \langle [h] \rangle \cong \mathbb{Z}^s / (F\mathbb{Z}^s + \mathbb{Z}h).$$

The group on the right has size $e_1 \cdots e_s$ by Smith normal form, while $|G_{\alpha}| = D$. Their ratio is the order of the cyclic subgroup, proving (7). The second formula is the coordinate divisibility condition after applying U . No determinant ratio is used without its lattice-index justification.

Since $h_l = (j)_{j \in H}$ once $l \geq \max H$, all larger carriers induce the same translation on this level. If $H = \emptyset$, the empty determinant and empty Smith product are one; the vacuum component is a singleton with $q = 1$.

3 Fixed points, aggregation, and primitive cycles

Translation by an element of order q on a finite group has a fixed point at iterate n precisely when $q \mid n$, in which case every point is fixed. Therefore one component satisfies

$$\# \text{Fix}(T_l^n | G_{\alpha}) = D_{\alpha} \mathbf{1}_{q_{\alpha, l} | n}. \quad (8)$$

After multiplying by the number of copies, the entire soliton level has

$$F_{m, l}(n) = \sum_{\alpha} c(\alpha) D_{\alpha} \mathbf{1}_{q_{\alpha, l} | n}. \quad (9)$$

Because the action variable partitions the binary words,

$$F_{L, M, l}(n) = \sum_{m: \sum j m_j = M} F_{m, l}(n), \quad F_{L, l}^+(n) = \sum_{M=0}^{\lfloor L/2 \rfloor} F_{L, M, l}(n), \quad (10)$$

with state totals $\binom{L}{M}$ and $\sum_{M \leq L/2} \binom{L}{M}$, respectively. Thus “all-state” means an exact finite aggregation, not a sampled orbit list.

Let P_n be the number of points of exact period n and $C_n = P_n / n$ the number of primitive cycles. Möbius inversion gives

$$P_n = \sum_{d | n} \mu(n/d) F(d), \quad C_n = P_n / n. \quad (11)$$

Equivalently, one component contributes $D_{\alpha} / q_{\alpha, l}$ cycles of length $q_{\alpha, l}$, multiplied by $c(\alpha)$. This proves integrality and completeness without a periodic-orbit cutoff.

4 Zeta and the finite Koopman determinant

For any one of the aggregations above, let C_q denote its number of length- q cycles. Equations (8)–(11) give

$$\zeta_{T_l}(z) = \exp\left(\sum_{n \geq 1} \frac{F(n)}{n} z^n\right) = \prod_q (1 - z^q)^{-C_q}. \quad (12)$$

On $\ell^2(X)$ with counting measure, $U_l f = f \circ T_l$ is a canonical finite unitary permutation. A q -cycle has characteristic polynomial $x^q - 1$, hence

$$\det(I - zU_l) = \prod_q (1 - z^q)^{C_q} = \zeta_{T_l}(z)^{-1}. \quad (13)$$

This is the ordinary determinant of the same-clock finite operator. It is neither a regularized infinite-volume determinant nor a Hilbert–Pólya construction.

5 Boundary closure

At half filling $p_N = 0$. In (5), all binomials for that coordinate equal one, so Möbius cancellation leaves only $\alpha_N = m_N$ with value one. Thus the component theorem remains finite at $L = 2M$. At $M = 0$, the one-point vacuum is fixed by every T_l . At $l \geq \max H$, the vector h_l saturates. These calculations close the three singular-looking edges without changing the clock or deleting states.

6 Route-A boundary

The exact cycles have no intrinsic prime labels, logarithmic periods, or source stability weights. The finite zeta therefore remains a source identity rather than a target divisor theorem. The final round adds the independent certificate and integrity audit.

References

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- [2] T. Takagi, “Level Set Structure of an Integrable Cellular Automaton,” *SIGMA* 6 (2010), 027, arXiv:0906.1410.